

IT2120 - Probability and Statistics

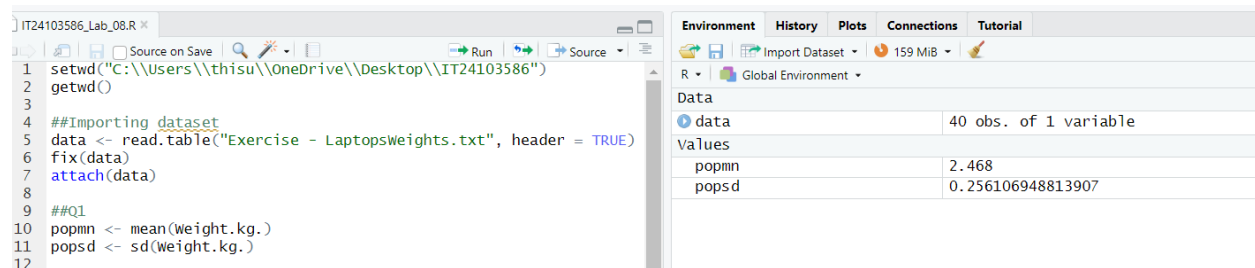
Lab Sheet 07

IT24103586 – Herath H.M.T.P

Exercise

Instructions: Create a folder in your desktop with your registration number (Eg: "IT....."). You need to save the R script file and take screenshots of the command prompt with answers and save it in a word document inside the folder. Save both R script file and word document with your registration number (Eg: "IT....."). After you finish the exercise, zip the folder and upload the zip file to the submission link.

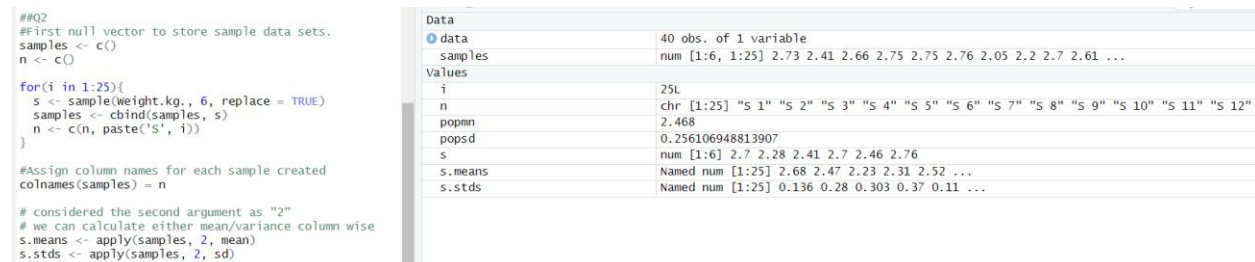
1. Calculate the population mean and population standard deviation of the laptop bag weights.



```
1 setwd("C:\\Users\\thisu\\OneDrive\\Desktop\\IT24103586")
2 getwd()
3
4 ##Importing dataset
5 data <- read.table("Exercise - LaptopsWeights.txt", header = TRUE)
6 fix(data)
7 attach(data)
8
9 ##Q1
10 popmn <- mean(weight.kg.)
11 popsd <- sd(weight.kg.)
12
```

Environment	
data	40 obs. of 1 variable
popmn	2.468
popsd	0.256106948813907

2. Draw 25 random samples of size 6 (with replacement) and calculate the sample mean and sample standard deviation for each sample.



```
##Q2
#First null vector to store sample data sets.
samples <- c()
n <- c()

for(i in 1:25){
  s <- sample(weight.kg., 6, replace = TRUE)
  samples <- cbind(samples, s)
  n <- c(n, paste('S', i))
}

#Assign column names for each sample created
colnames(samples) = n

# considered the second argument as "2"
# we can calculate either mean/variance column wise
s.means <- apply(samples, 2, mean)
s.stds <- apply(samples, 2, sd)
```

Data	
data	40 obs. of 1 variable
samples	num [1:6, 1:25] 2.73 2.41 2.66 2.75 2.75 2.76 2.05 2.2 2.7 2.61 ...
n	chr [1:25] "S 1" "S 2" "S 3" "S 4" "S 5" "S 6" "S 7" "S 8" "S 9" "S 10" "S 11" "S 12" ...
popmn	2.468
popsd	0.256106948813907
s	num [1:6] 2.7 2.28 2.41 2.7 2.46 2.76
s.means	Named num [1:25] 2.68 2.47 2.23 2.31 2.52 ...
s.stds	Named num [1:25] 0.136 0.28 0.303 0.37 0.11 ...

3. Calculate the mean and standard deviation of the 25 sample means and state the relationship of them with true mean and true standard deviation.

<pre>##Q3 samplemean <- mean(s.means) samplestd <- sd(s.means) #compare the population mean and mean of sample means. popmn samplemean #compare the population std and std of sample means. truesd = popsd/6 samplestd</pre>	
R Global Environment	
Data	40 obs. of 1 variable
data	
samples	num [1:6, 1:25] 2.73 2.41 2.66 2.75 2.75 2.76 2.05 2.2 2.7 2.61 ...
Values	
i	25L
n	chr [1:25] "S 1" "S 2" "S 3" "S 4" "S 5" "S 6" "S 7" "S 8" "S 9" "S 10" "S 11" "S 12" .
popmn	2.468
popsd	0.256106948813907
s	num [1:6] 2.7 2.28 2.41 2.7 2.46 2.76
s.means	Named num [1:25] 2.68 2.47 2.23 2.31 2.52 ...
s.stds	Named num [1:25] 0.136 0.28 0.303 0.37 0.11 ...
samplemean	2.48006666666667
samplestd	0.108070707306734
truesd	0.0426844914689845

```
Console | Terminal x | Background Jobs x
R | R 4.5.1 | C:/Users/thisu/OneDrive/Desktop/IT24103586/
> ##Q3
> samplemean <- mean(s.means)
> samplestd <- sd(s.means)
>
> #compare the population mean and mean of sample means.
> popmn
[1] 2.468
> samplemean
[1] 2.480067
>
>
> #compare the population std and std of sample means.
> truesd = popsd/6
> samplestd
[1] 0.1080707
> |
```